

Highlights

DSN: Energy-Efficient EMG Signal Classification Method Enabling Real-Time Monitoring for Edge Healthcare

Zhenyu Zhang, Xianzhe Meng, Jianglan Wei, Mingjie Zeng, Zhigang Zeng

- Breaks the sEMG energy barrier by fusing SNN dynamics with HDC symbolic reasoning in a unified DSN framework.
- Eliminates backpropagation and heavy training through randomized spiking feature extraction and one-pass hyperdimensional encoding.
- Provides formal guarantees of robustness to noise, bit-flip faults and convergence through theoretical analysis of SNN and HDC modules.
- Surpasses GRU with $42.42\times$ lower energy and reaches 93.87% accuracy using only 20% of training data in three-class fatigue detection.